



Oncor Venus Switch to Sam Switch 345-kV Line - ERCOT Independent Review Study Scope

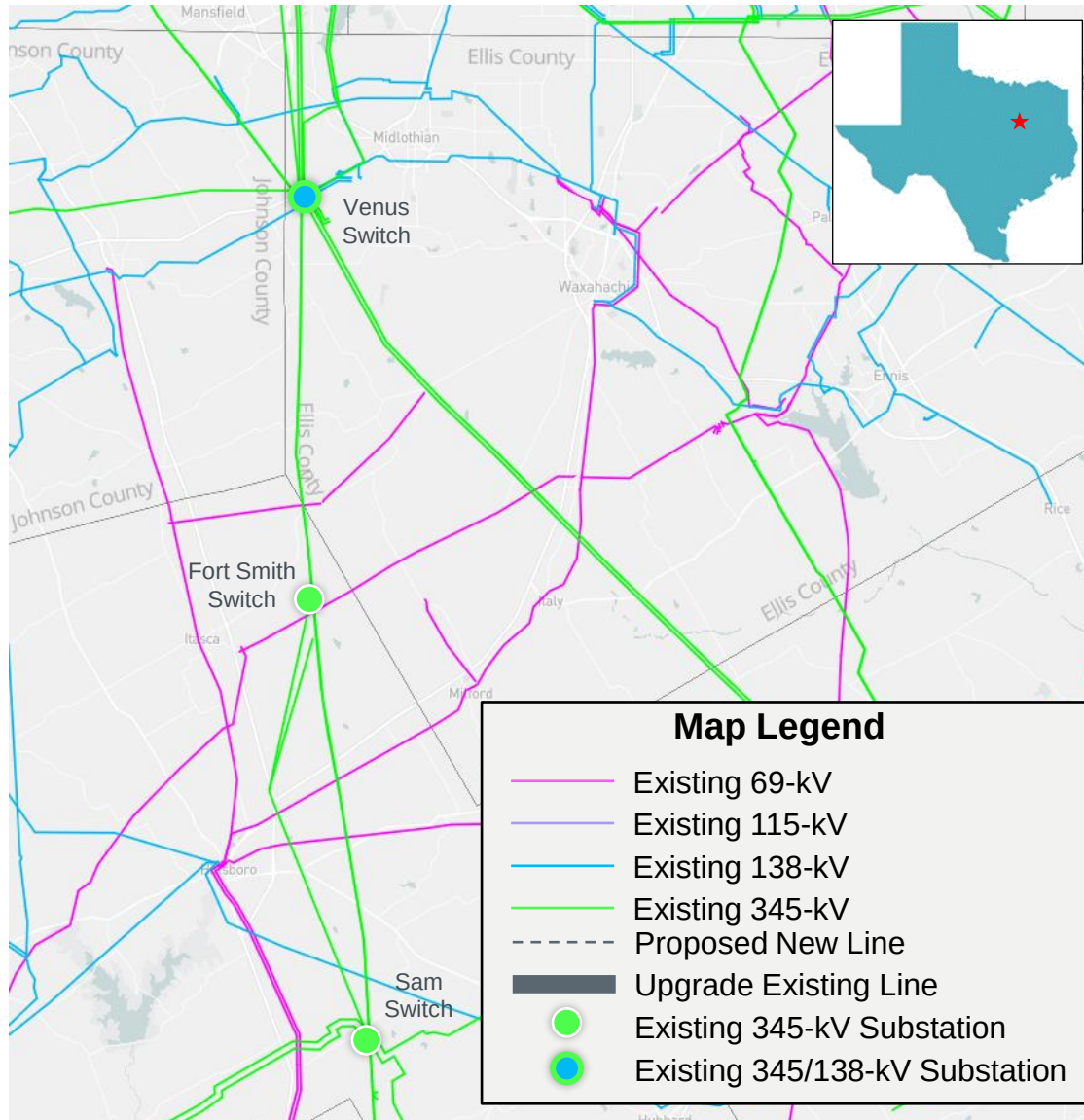
Sarah Gunasekera

RPG Meeting
August 13th, 2024

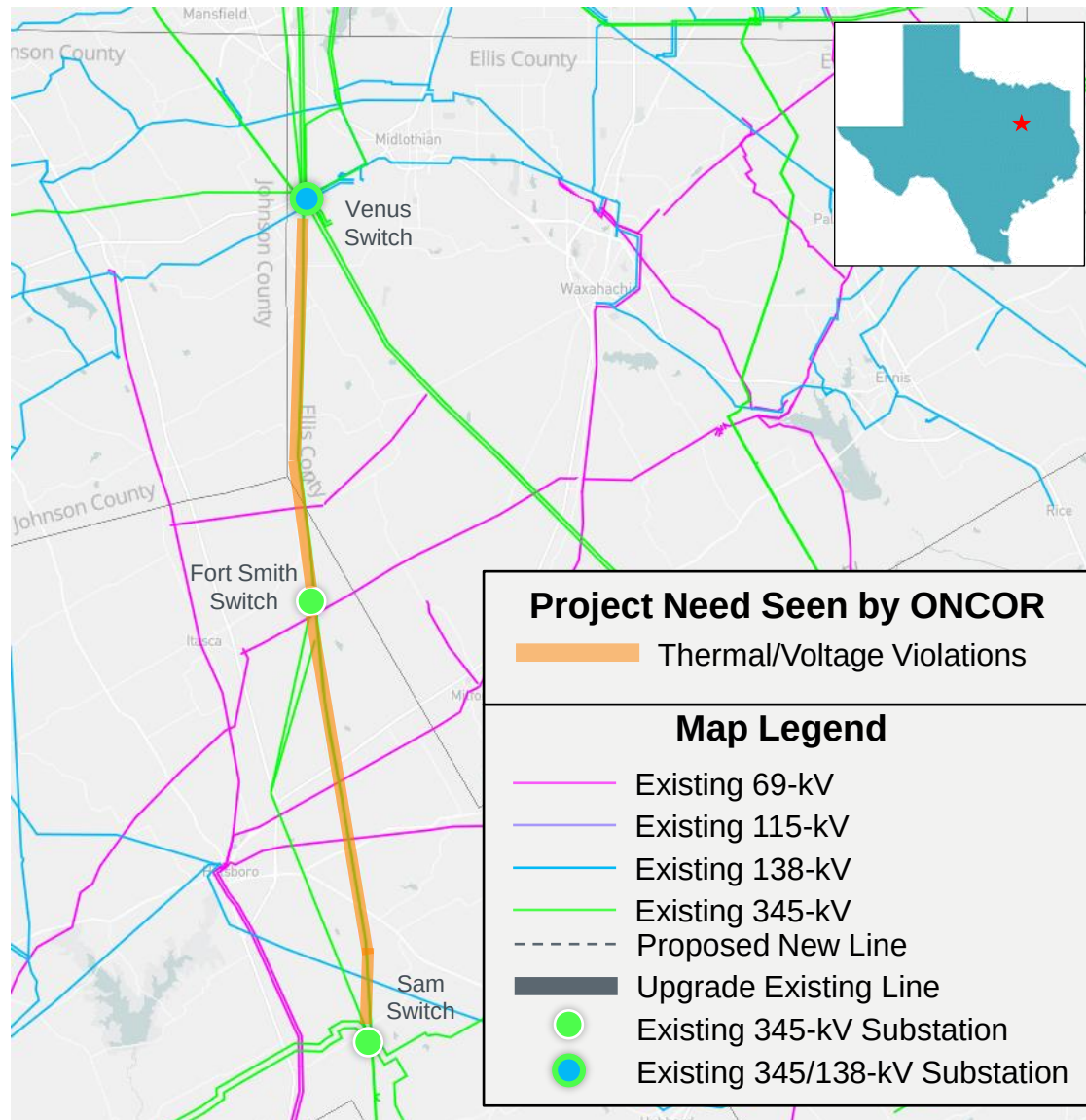
Introduction

- Oncor submitted the Venus Switch to Sam Switch 345-kV Line Project for Reginal Planning Group (RPG) review in June 2024
 - This Tier 1 project is estimated at \$118.9 million and will not require a Convenience and Necessity (CCN)
 - Estimated completion date is May 2026
 - Addresses post-contingency thermal overloads on the Venus Switch – Sam Switch double circuit 345-kV line
- This project is currently under ERCOT Independent Review (EIR)

Study Area Map



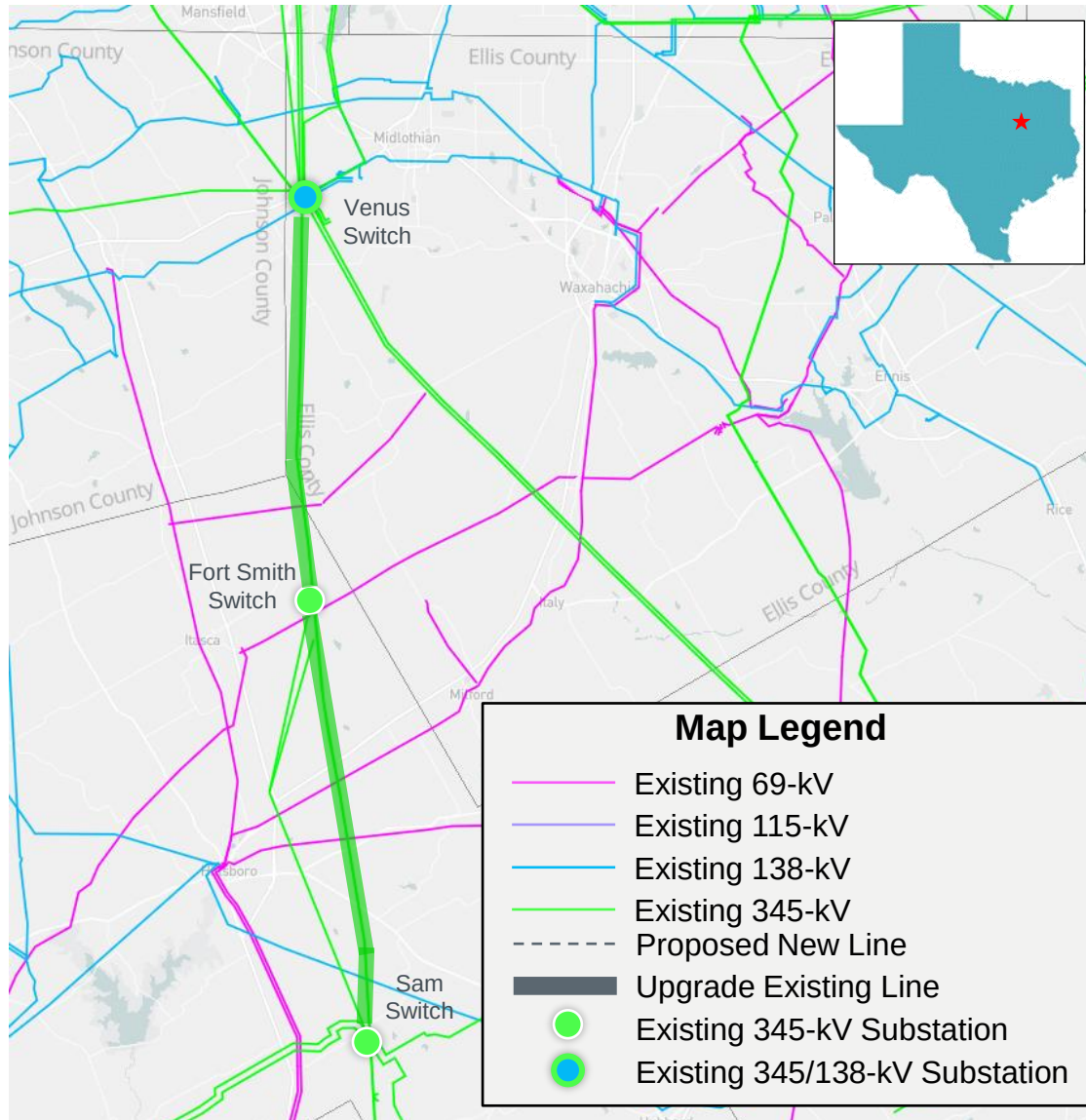
Study Area Map with Project Need seen by Oncor



Project Proposed by Oncor

- Rebuild existing Venus Switch – Fort Smith Switch 345-kV transmission line with normal and emergency ratings of 1912 MVA or greater, approximately 17.80-mile
- Rebuild existing Fort Smith Switch – Sam Switch 345-kV transmission line with normal and emergency ratings of 1792 MVA or greater, approximately 20.20-mile
- Rebuild existing Venus Switch – Sam Switch 345-kV transmission line with normal and emergency ratings of 1792 MVA or greater, approximately 38.0-mile

Proposed Project by Oncor



Study Assumptions – Base Case

- Study Area
 - North and North Central Weather Zones, focusing on transmission in Ellis and Hill Counties
 - Monitor surrounding counties that are electrically close to the area
- Steady-State Base Case
 - Final 2023 Regional Transmission Planning (RTP) 2026 summer peak case for North-North Central Weather Zones will be updated to construct the North-North Central (NNC) study base case posted in Market Information System (MIS)
 - Case: 2023RTP_2026_SUM_NNC_12222023
 - Link: <https://mis.ercot.com/secure/data-products/grid/regional-planning>

Study Assumption - Transmission

- Based on the Transmission Project and Information Tracking (TPIT) published on MIS in February 2024, projects within the study area with in-service dates prior to May 2026 were added to the study base case if not already modeled in the case
 - TPIT Link: <https://www.ercot.com/gridinfo/planning>
 - See Appendix A for the list of transmission projects added
- Transmission projects identified in the 2023 RTP in the study area that have not been approved by RPG will be removed from the study base case
 - See Appendix B for the list transmission projects that have been backed out

Study Assumptions – Generation

- New generation that met Planning Guide Section 6.9(1) condition with Commercial Operation Date (COD) before the May 2026 in the study area at the time of the study, but not already modeled in the RTP cases, will be added to the case based on Generator Interconnection Status (GIS) report published in MIS in July 2024
 - Link: <https://www.ercot.com/gridinfo/resource>
 - See Appendix C for the list of generation projects added to the case
- All generation will be dispatched consistent with the 2024 RTP methodology
- All recent retired/indefinitely mothballed units will be reviewed and opened (turned off), if not already reflected in the 2023 RTP Final cases

Study Assumptions – Load & Reserve

- Loads in study area
 - Load level in the study area will be maintained consistent with the 2023 RTP case
 - Newly approved loads in the study area will be added to the study base case
- Reserve
 - Load outside of NNC Weather Zones may be adjusted to maintain the reserve consistent with the 2023 RTP

Contingencies & Criteria

- Contingencies for study region
 - NERC TPL-001-5.1 and ERCOT Planning Criteria
 - Link: <http://www.ercot.com/mktrules/guides/planning/current>
 - P0 (System Intact)
 - P1, P2-1, P7 (N-1 conditions)
 - P2-2, P2-3, P4, and P5 (345-kV only)
 - P3: G-1+N-1 (G-1: Midlothian ANP, Compadre Solar, and Sunvalley Solar and Storage)
 - P6: X-1+N-1 (X-1: 345/138-kV Sherry Switch, and Everman Switch, Lake Creek SES)
- Criteria
 - Monitor all 60 kV and above busses, transmission lines, and transformers in the study region (excluding generator step-up transformers)
 - Thermal
 - Use Rate A for normal conditions
 - Use Rate B for emergency conditions
 - Voltage
 - Voltages exceeding their pre-contingency and post-contingency limits
 - Voltage deviations exceeding 8% on non-radial load buses

Study Procedure

- Need Analysis
 - The reliability analysis will be performed to identify the need to serve the projected area load using the study base case
- Project Evaluation
 - Project alternatives will be tested to satisfy the NERC and ERCOT reliability requirements
 - ERCOT may also perform the following studies
 - Planned maintenance outage
 - Long-term Load Serving Capability Assessment
- Generation Addition and Load Scaling Sensitivity Analyses
 - Planning Guide Section 3.1.3(4)
- Subsynchronous Resonance (SSR) Assessment
 - Nodal Protocol Section 3.22.1.3(2)
- Congestion Analysis
 - Congestion analysis may be performed based on the recommended transmission upgrades to ensure that the identified transmission upgrades do not result in new congestion within the study area

Deliverables

- Tentative Timelines
 - Status updates at the future RPG meetings
 - Final Recommendation – Q4 2024

Thank you!



Stakeholder comments also welcomed through:

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Appendix A – Transmission Projects Added

TPIT/RPG No	Project Name	Tier	Project ISD	TSP	County(s)
24RPG019	Vineyard Switch to Cypress Waters 138-kV Circuit Addition Project	Tier 2	May-26	ONCOR	Dallas, Tarrant
23RPG033	Watermill to Seagoville 138 kV Line Project	Tier 3	Dec-25	ONCOR	Dallas
23RPG020	Hackberry Switch to DFW D East 2 138-kV Double-Circuit Line Section Project	Tier 3	Dec-25	ONCOR	Dallas
23RPG018	Arlington Reliability Enhancement Project	Tier 2	May-26	ONCOR	Tarrant, Dallas
23RPG006	North Lake 138 kV Switch Rebuild	Tier 4	May-23	ONCOR	Dallas
66218A	Hillsboro - Italy 69 kV Line	Tier 4	Oct-23	ONCOR	Ellis
66218B	Hillsboro - Italy 69 kV Line	Tier 4	Dec-25	ONCOR	Ellis
60094	Convert Waco East - Elm Mott 69 kV Line to 138 kV	Tier 4	May-24	ONCOR	McLennan
73443	Utilize Melton POI via Navarro 345 kV Switch for Project Lefty	Tier 4	May-24	ONCOR	Navarro
66216	Upgrade and convert Waco West - Temple 69 kV Line to 138 kV	Tier 4	Jun-24	ONCOR	McLennan, Bell
62666	Upgrade and convert McGregor - Waco West Line	Tier 4	Dec-24	ONCOR	McLennan
80550	Central Park 138 kV Switch	Tier 4	Dec-24	ONCOR	McLennan
82826	Sunflower 138 kV Switch	Tier 4	May-25	ONCOR	McLennan
78167	Add 2nd autotransformer at Trumbull	Tier 4	Nov-25	BEPC	Ellis
71903	Establish Launch Pad 138 kV Switch	Tier 4	Dec-25	ONCOR	McLennan
71136	Waxahachie-Waxahachie OCF 69 kV Line Rebuild	Tier 4	May-25	ONCOR	Dallas, Ellis

Appendix B – Transmission Backed Out

RTP Project ID	Project Name	TSP	County(s)
2023-NC6	Telico Area Upgrades	ONCOR	Ellis
2023-NC7	Four Brothers Switch -Tradinghouse - Outlaw - Lake Hall Switch - Sam Switch Area Improvements	ONCOR	Mclennan, Ellis
2023-NC13	Hillboro 138-kV Area Upgrades	BEPC	Hill
2023-NC16	Sardis Area 138-kV Line Upgrades	ONCOR	Ellis
2023-NC19	Venus - Fort Smith - Sam Switch Double Circuit 345-kV Line Upgrades and Venus Kemp Ranch 345/138-kV Transformer Addition	ONCOR	Ellis
2023-NC23	Venus - Navarro - Jewett Area 345-kV Line Upgrades	ONCOR	Ellis
2023-NC35	Navarro (3478) - Haney (213) - Hubbard (3515) 138-kV Line Upgrade	ONCOR	Navarro
2023-NC37	Hillboro 138/69-kV Transformer Upgrade	ONCOR	Hill
2023-NC42	Waxahachie Area 69-kV and 138-kV Line Upgrades	ONCOR	Ellis
2023-NC45	Cleburne Switch (2279) to Keene (2294) to Alvarado (2297) to Griffith (1905) to Railport (442) to Venus (1908) 138-kV Line Upgrades	BEPC, ONCOR	Johnson
2023-NC62	Whitney 345/138-kV Transformer Upgrade	BEPC	Hill

Appendix C – Generation Added

GINR	Project Name	Fuel	Project COD	Capacity (MW)	County
21INR0379	Ash Creek Solar	SOL	01/31/2025	417.74	Hill
23INR0030	Langer Solar	SOL	03/01/2027	249.8	Bosque
23INR0195	Desert Willow BESS	OTH	02/03/2025	154.4	Ellis
23INR0349	Tokio Solar	SOL	08/25/2025	175.72	McLennan
24INR0023	Compadre Solar	SOL	12/25/2024	406.14	Hill
24INR0038	SP Jaguar Solar	SOL	06/01/2026	300	McLennan
24INR0039	SP Jaguar BESS	OTH	06/30/2025	300	McLennan
24INR0138	Midpoint Storage	OTH	08/30/2025	51.26	Hill
24INR0139	Midpoint Solar	SOL	08/30/2025	99.8	Hill
24INR0140	Gaia Storage	OTH	07/31/2025	76.83	Navarro
24INR0141	Gaia Solar	SOL	07/31/2025	152.65	Navarro